Temperature Monitoring and Thermal Control System

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Temperature Monitoring and Thermal Control System
Embedded Internship Assignment
Agenthix
Submitted by:
Vaka Harshini
Date:
14 March 2026
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1. Introduction

Temperature monitoring is an essential function in many embedded systems such as industrial automation, environmental monitoring, and electronic thermal management. Excess temperature can damage electronic components or reduce system performance.

This project implements a temperature monitoring and thermal control system using a microcontroller. The system reads temperature from a sensor, displays the measured value on an LCD display, and automatically controls a cooling fan to maintain safe operating conditions.

2. Objective

The objective of this project is to design and implement a temperature monitoring and control system using a microcontroller. The system continuously measures temperature and displays the value on an LCD. If the temperature exceeds a defined threshold, a fan is automatically activated to reduce the temperature and maintain it within the safe range of 25°C to 50°C.

3. System Architecture

Temperature Sensor → Microcontroller → LCD Display

↓

Fan Driver

↓

Cooling Fan

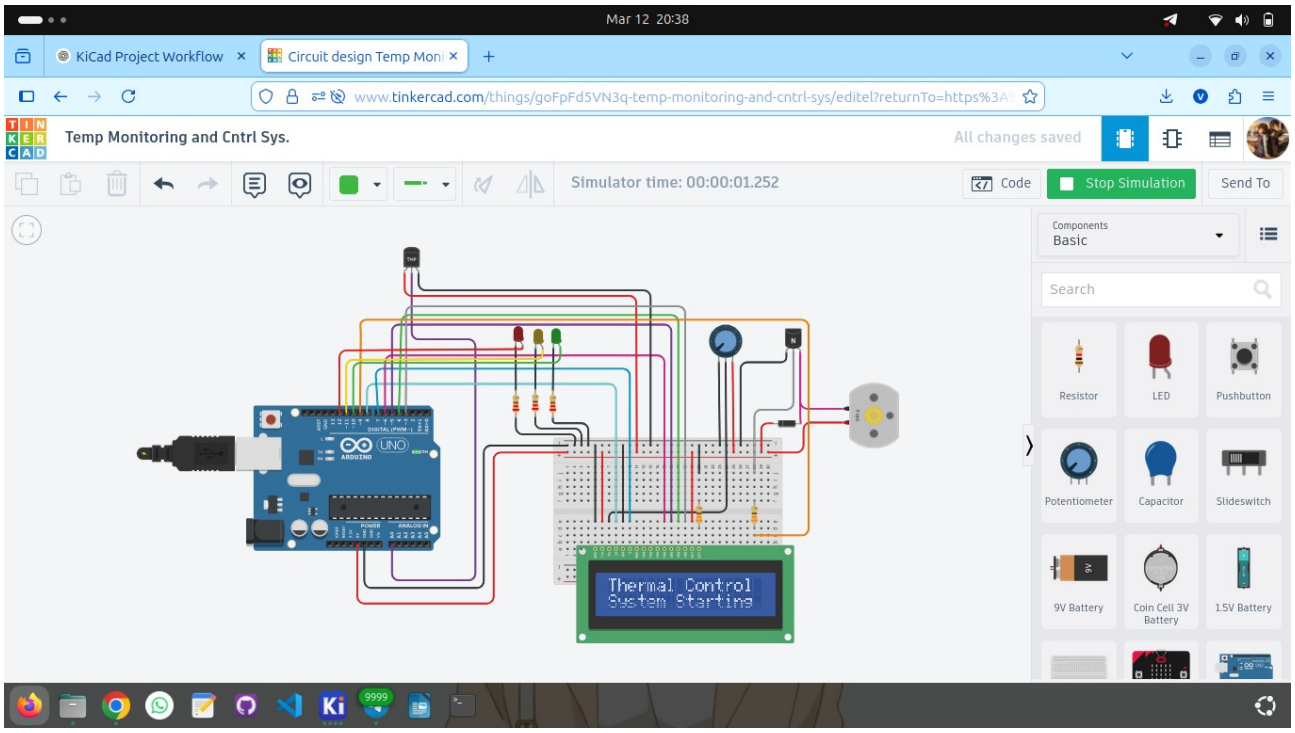
The temperature sensor measures the ambient temperature and sends an analog signal to the microcontroller. The microcontroller processes this signal using its ADC module and converts it into a temperature value. The measured temperature is displayed on the LCD display. If the temperature exceeds the predefined threshold, the microcontroller activates the cooling fan through a transistor driver circuit.

4. Hardware Components

S.NO	Component	Purpose
1	Microcontroller	Processes temperature data
2	Temperature Sensor	Measures temperature
3	16x2 LCD Display	Displays temperature value
4	Cooling Fan	Reduces temperature
5	Transistor	Drives the fan motor
6	Flyback Diode	Protects circuit from back EMF
7	Resistors	Circuit stability

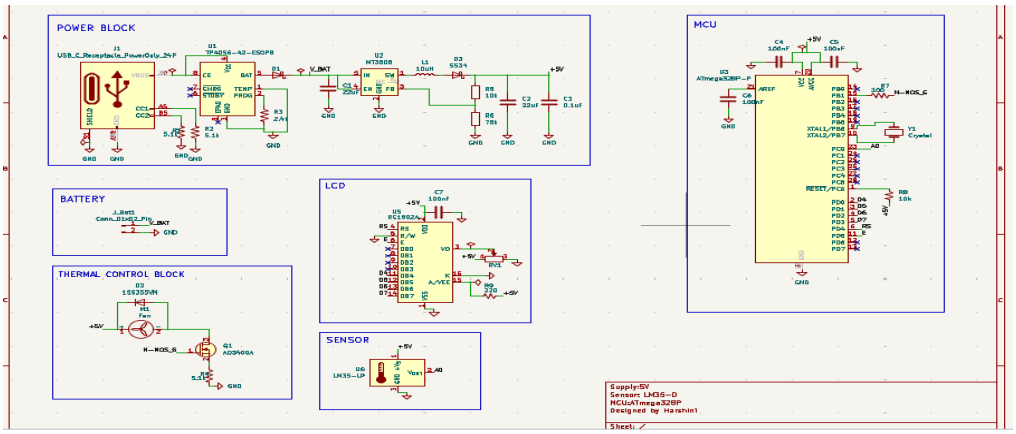
5. Circuit Diagram

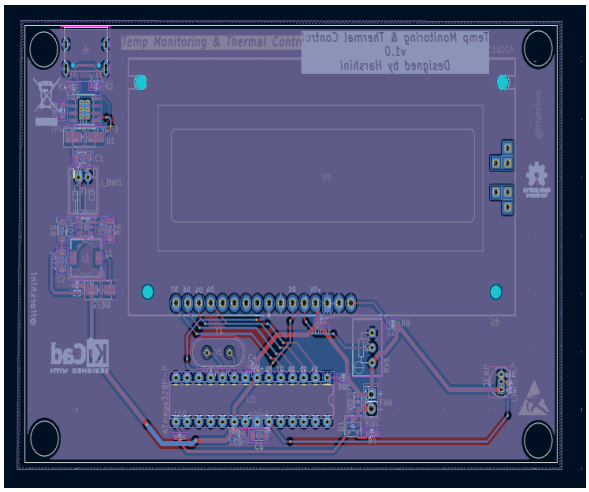
Temperature Monitoring System Circuit Diagram

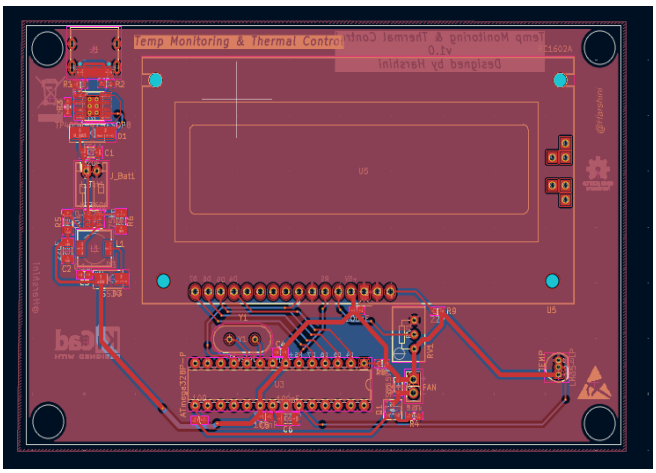


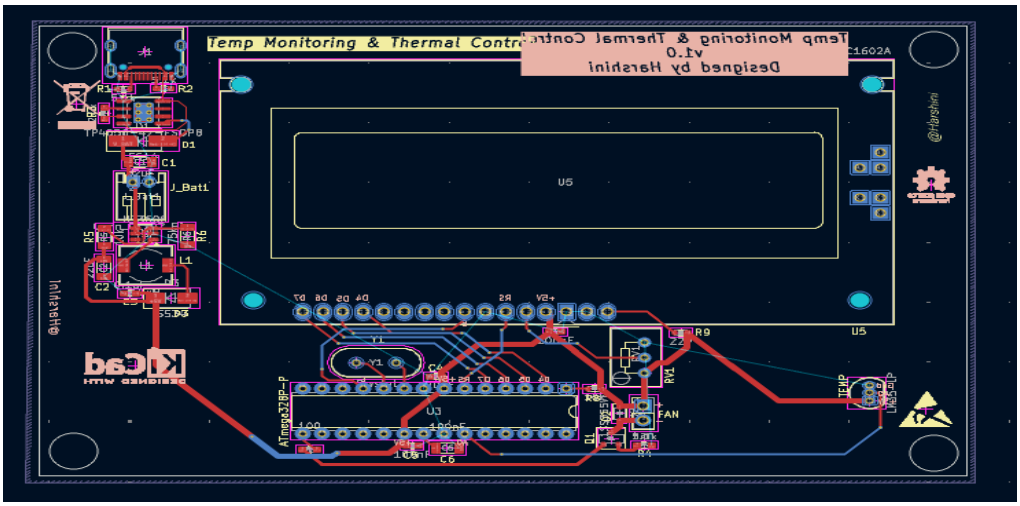
6. PCB Design

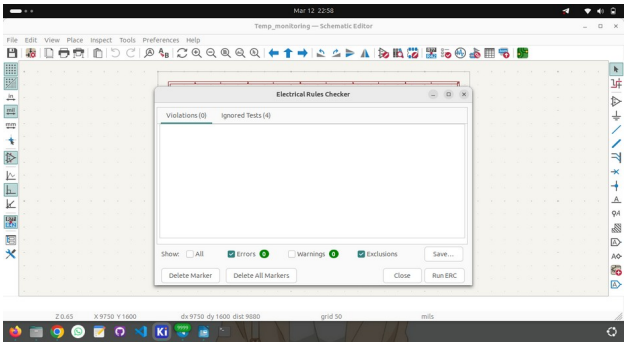
A custom PCB layout was designed using KiCad to demonstrate practical hardware implementation. The PCB includes footprints for the microcontroller, LCD display connector, temperature sensor, fan driver circuit, and power supply section. Proper routing techniques and component placement were used to ensure efficient signal flow and stable power distribution.


POWER BLOCK
BATTERY
THERMAL CONTROL BLOCK
LCD
SENSOR
MCU
Supply:5V
Sensor: LM35-D
MCU:ATmega328P
Designed by Harshini
Sheet: /






Temp Monitoring & Thermal Contro
Temp Monitoring & Thermal Control
v1.0
Designed by Harshini
C1602A
U5
J_Batt
D1
C1
C2
L1
R1
R2
R9
RV1
U3
ATmega328P-P
FAN
TEMP
LM35-LP
@Harshini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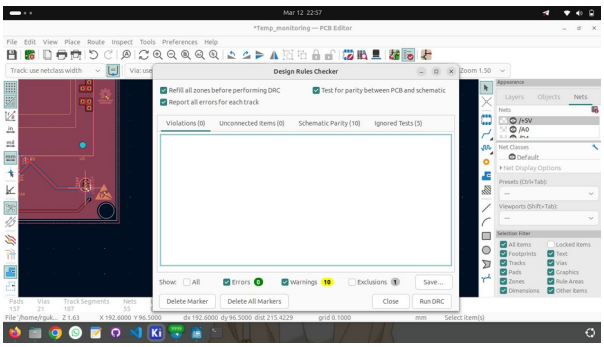
Electrical Rules Checker
Violations (0)
Ignored Tests (4)
Delete Marker
Delete All Markers
Close
Run ERC


*Temp_monitoring — PCB Editor
Design Rules Checker
Violations (0)
Unconnected items (0)
Schematic Parity (10)
Ignored Tests (5)
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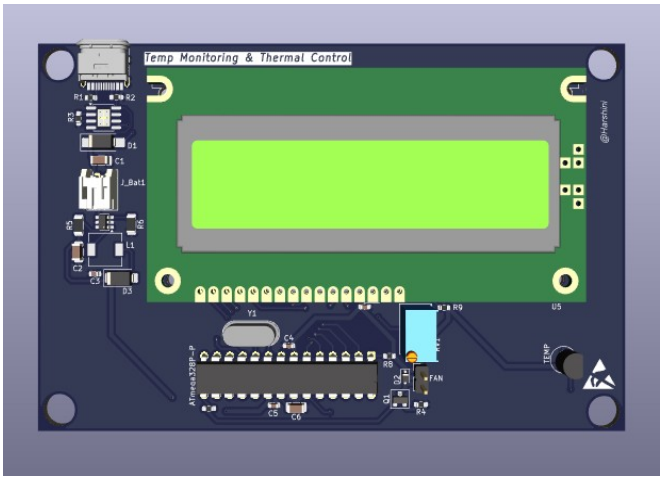
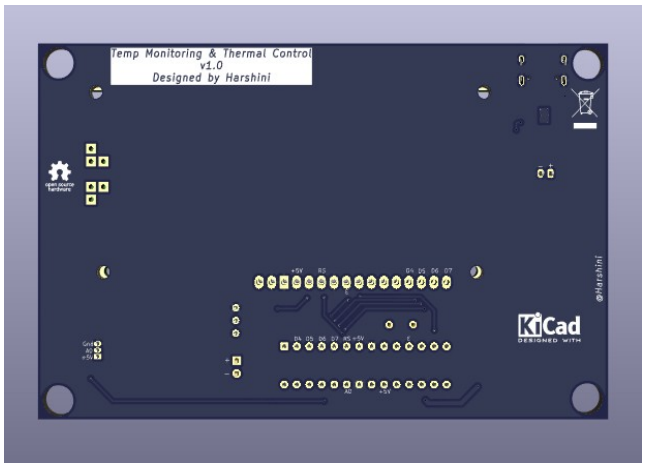
Delete All Markers
Close
Run DRC





7. Software Implementation

Logic:

1. Read temperature sensor
2. Convert analog voltage to temperature
3. Display temperature on LCD
4. Check threshold value
5. Turn fan ON/OFF
6. Display system status

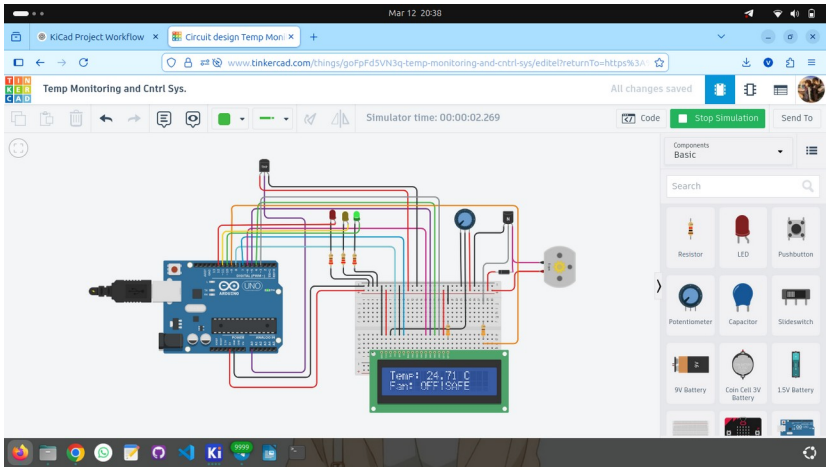
i.e.,

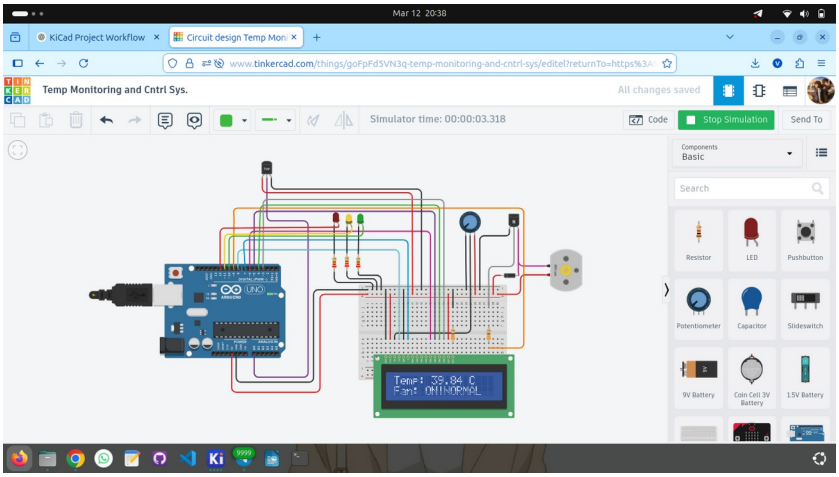
Temp < 25°C → Fan OFF

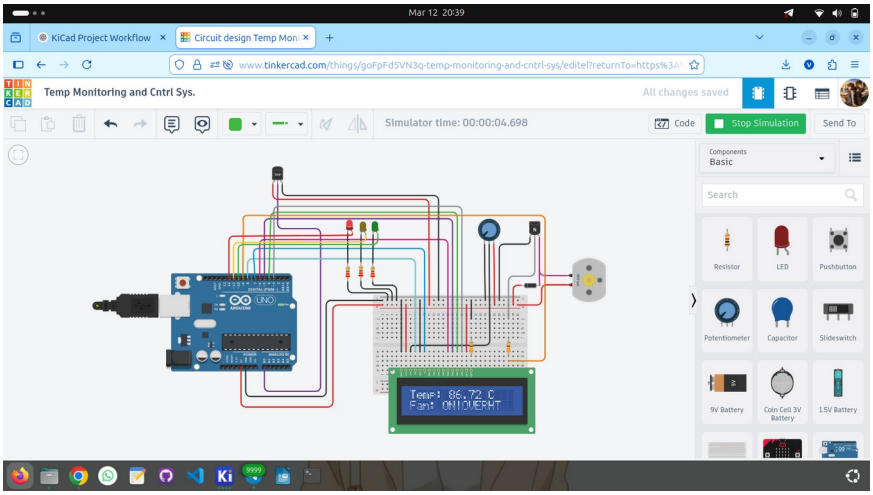
25°C – 50°C → Fan ON

> 50°C → OVERHEAT warning

8. Simulation Results







Temp: 21.71°C → Fan OFF |Safe (Green LED ON)

Temp: 39.84°C → Fan ON | Normal ,(Yellow LED ON)

Temp: 86.72°C → OVERHEAT Warning ,(Red LED ON)

9. Results

The system successfully monitors temperature and controls the cooling fan automatically. The LCD display shows real-time temperature values and system status. The simulation verifies correct fan activation and overheating alerts based on temperature thresholds.

10. Conclusion

This project demonstrates a practical embedded system for temperature monitoring and thermal control. The system successfully integrates sensor interfacing, microcontroller processing, LCD display, and actuator control. Additionally, a PCB layout was designed to demonstrate readiness for hardware implementation. The project showcases both software and hardware aspects of embedded system design.

11. Future Improvements

Future improvements may include:

• PWM fan speed control

12. Verification Link

Click here to open Tinkercad link

Click here to open PCB kicad files

Click here to get the code